CHAPTER 05

Alcohols

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

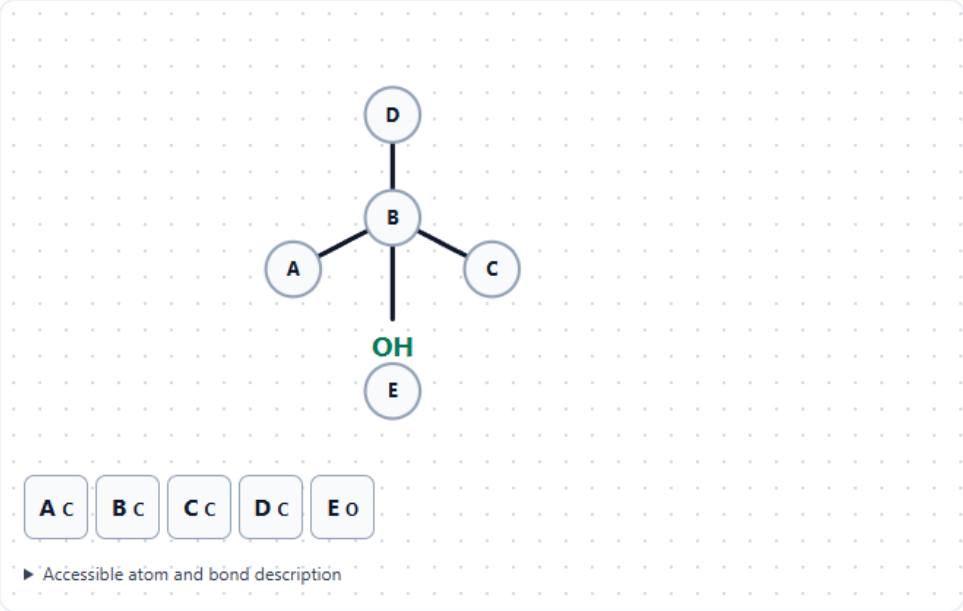
Source snapshot SHA-256: abaec75d990db6ff0ddd7d875947d6667a6be2f1a7584af2577b306ee8379625

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

classify-alcohol | Independent exercise

Screenshot: classify-alcohol-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=25.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=160.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

classify-alcohol:stage-1 | Select the OH-bearing carbon.

Answer type: atoms

classify-alcohol:stage-2 | How many carbon neighbors does it have?

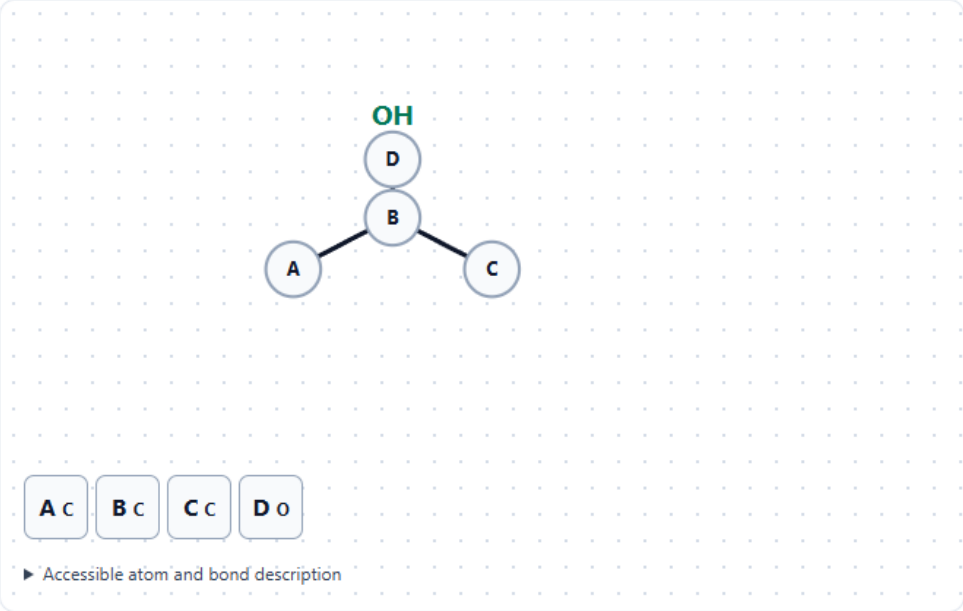
Answer type: number | unit: carbon neighbors

classify-alcohol:stage-3 | Classify this alcohol: primary, secondary, or tertiary.

Answer type: text

oxidize-secondary | Independent exercise

Screenshot: oxidize-secondary-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

oxidize-secondary:stage-1 | Select the carbon that changes oxidation state.

Answer type: atoms

oxidize-secondary:stage-2 | Ordinary oxidation produces which functional group?

Answer type: choice

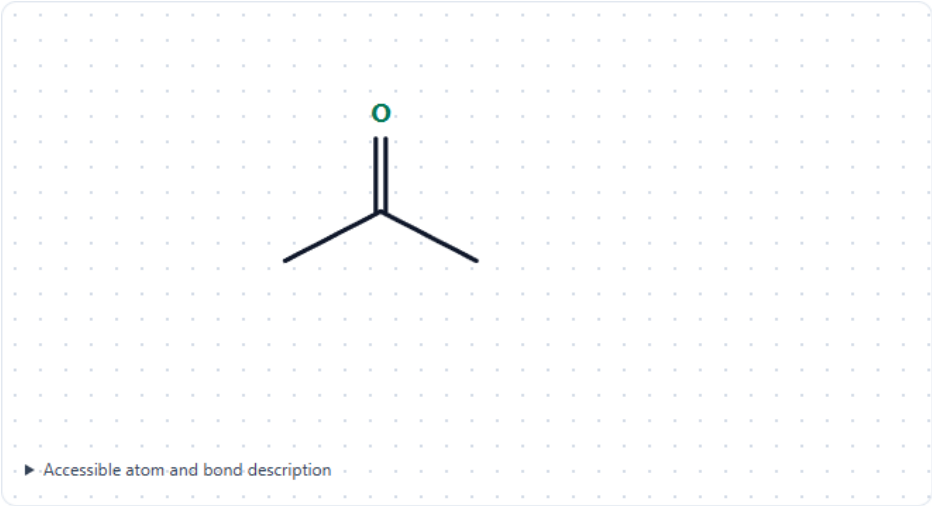
Option 0 (A): Ketone

Option 1 (B): Aldehyde

Option 2 (C): Carboxylic acid without C–C cleavage

This stage uses a different molecule (connectivity shown).

Screenshot: oxidize-secondary-stage-3-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

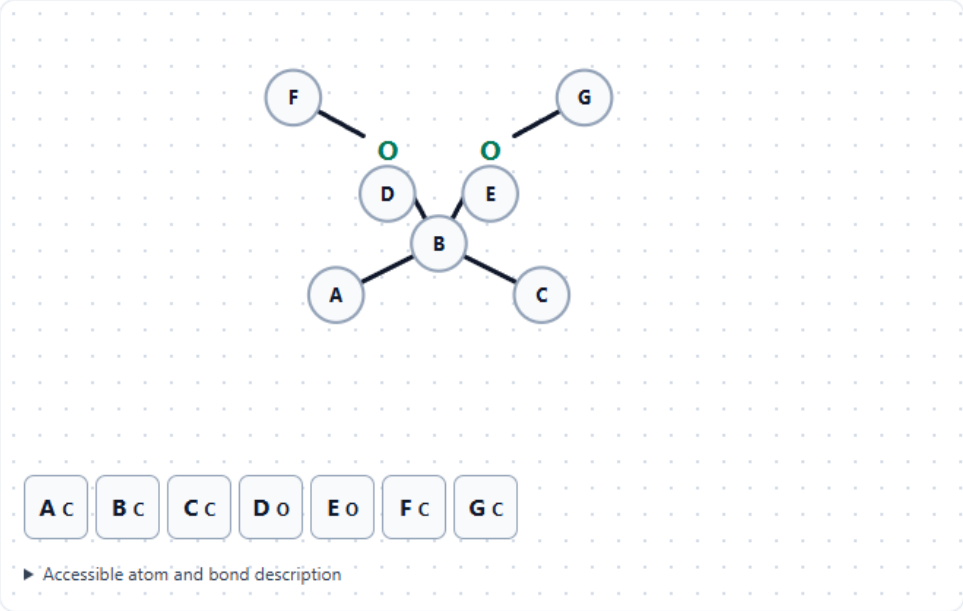
Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

oxidize-secondary:stage-3 | Name the three-carbon product.

Answer type: text

protection | Independent exercise

Screenshot: protection-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=75.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=135.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=195.00, y=130.00; charge=0; explicit H override=none

3 (D): O; x=105.00, y=45.00; charge=0; explicit H override=none

4 (E): O; x=165.00, y=45.00; charge=0; explicit H override=none

5 (F): C; x=50.00, y=15.00; charge=0; explicit H override=none

6 (G): C; x=220.00, y=15.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1], [3, 5, 1], [4, 6, 1]]

protection:stage-1 | Select the carbon bearing both OR groups.

Answer type: atoms

protection:stage-2 | Which conditions ordinarily regenerate the ketone?

Answer type: choice

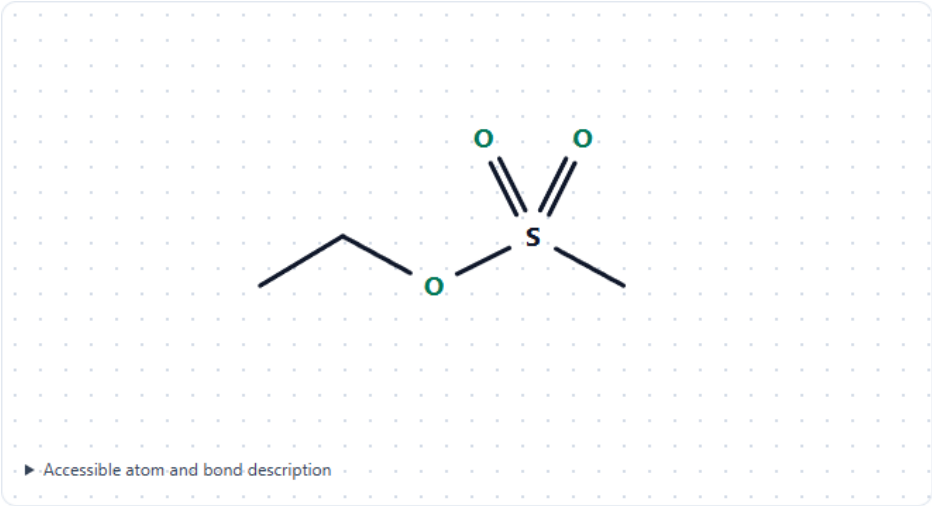
Option 0 (A): Aqueous acid

Option 1 (B): Dry neutral solvent alone

Option 2 (C): Base alone under ordinary conditions

This stage uses a different molecule (connectivity shown).

Screenshot: protection-stage-3-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=35.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=85.00, y=100.00; charge=0; explicit H override=none

2 (C): O; x=140.00, y=130.00; charge=0; explicit H override=none

3 (D): S; x=200.00, y=100.00; charge=0; explicit H override=none

4 (E): C; x=255.00, y=130.00; charge=0; explicit H override=none

5 (F): O; x=170.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=230.00, y=40.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [3, 5, 2], [3, 6, 2]]

protection:stage-3 | Making an alkyl sulfonate from ROH (mesylate shown; tosylate analogous) directly causes…

Answer type: choice

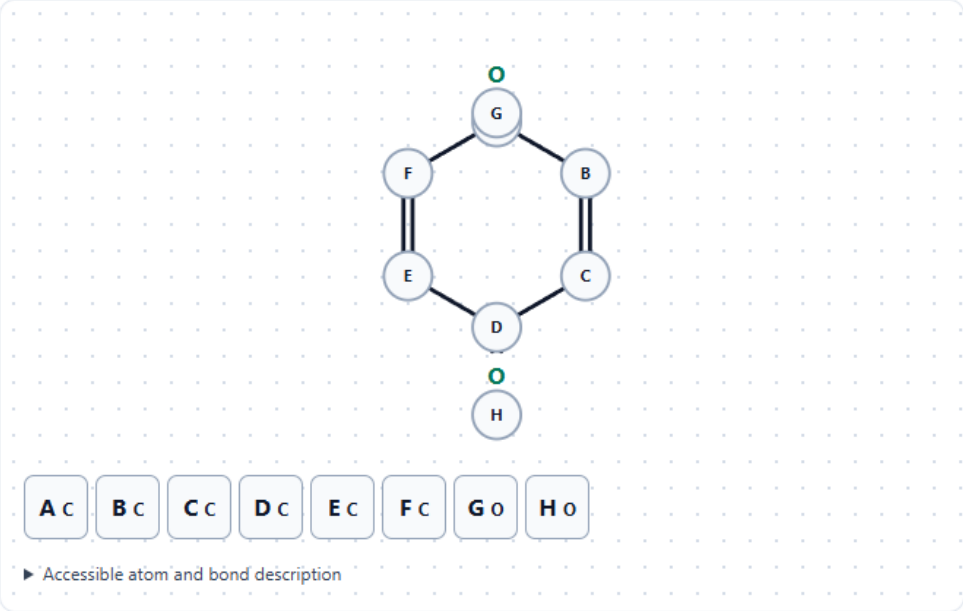
Option 0 (A): Retention of the C–O framework

Option 1 (B): SN2 inversion at carbon in every case

Option 2 (C): Cleavage of the carbon skeleton

quinone-redox | Independent exercise

Screenshot: quinone-redox-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): O; x=170.00, y=5.00; charge=0; explicit H override=none

7 (H): O; x=170.00, y=205.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 2], [3, 7, 2]]

quinone-redox:stage-1 | Select both carbonyl oxygen atoms.

Answer type: atoms

quinone-redox:stage-2 | Quinone → hydroquinone is…

Answer type: choice

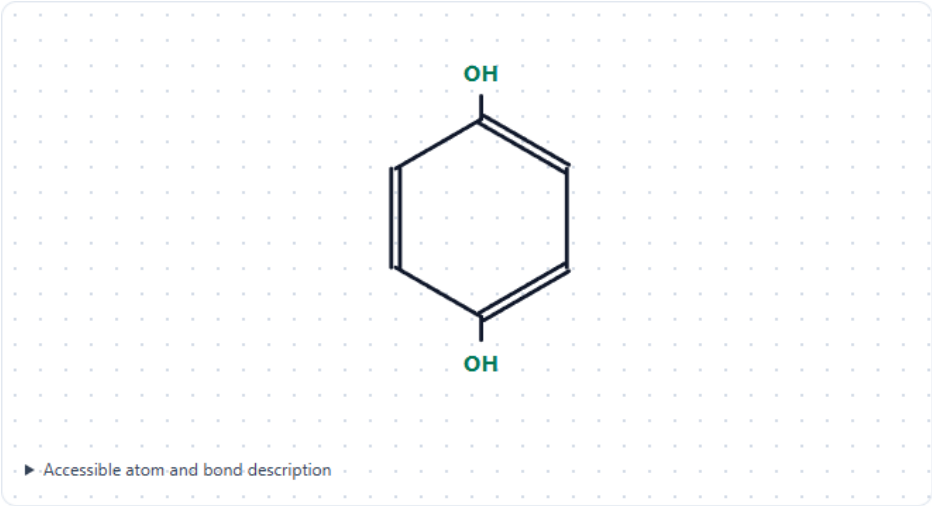
Option 0 (A): Reduction

Option 1 (B): Oxidation

Option 2 (C): Only a conformational change

This stage uses a different molecule (connectivity shown).

Screenshot: quinone-redox-stage-3-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): O; x=170.00, y=5.00; charge=0; explicit H override=none

7 (H): O; x=170.00, y=205.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [3, 7, 1]]

quinone-redox:stage-3 | The reduced product is shown. How many electrons are accepted per quinone in the full quinone → hydroquinone conversion?

Answer type: number | unit: electrons